

ABHINANDAN KAUSTHUB

Structural Integrity & Durability Engineer | Fatigue & Damage Tolerance | Fracture Mechanics

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PROFESSIONAL SUMMARY

Structural Integrity and Durability Engineer specializing in fatigue and fracture mechanics, structural life prediction, and composite structures. Skilled in fatigue crack growth modeling, damage tolerance analysis, and computational engineering using Python and MATLAB. Experienced in composite laminate analysis using Classical Lamination Theory (CLT), CFD simulation (ANSYS Fluent), and experimental mechanics for aerospace structural applications.

TECHNICAL SKILLS

Fatigue & Fracture: Crack Growth Modeling, Damage Tolerance Analysis, Structural Life Prediction, Paris Law, Walker Model, Forman Equation, Basquin S-N Curves, Coffin-Manson Strain-Life, Miner's Rule Damage Accumulation, Rainflow Cycle Counting, Mean Stress Correction

Structural Analysis: Finite Element Analysis (FEA), ANSYS, Composite Structures, Classical Lamination Theory (CLT), ABD Stiffness Matrix, Ply Stress Analysis, Thermal-Structural Analysis, Progressive Ply Failure, Stress/Strain Analysis

Failure Analysis: Maximum Stress Criterion, Tsai-Hill Criterion, Tsai-Wu Criterion, FMEA, Statistical Process Control (SPC)

CFD & Simulation: ANSYS Fluent, Compressible Flow, Navier-Stokes Equations, k- ω Turbulence Modeling, Hypersonic Flow Analysis, Modal Analysis

Programming: Python, MATLAB, NumPy, SciPy, Scientific Computing, Monte Carlo Simulation

CAD & Standards: CATIA, SolidWorks, Parametric Modeling, Surface Modeling, GD&T, Engineering Drawings

PLM: PTC Windchill/Creo, PDMLink, eBOM/mBOM Management, ECR/ECN Workflows, Product Lifecycle Management (In Progress)

ENGINEERING EXPERIENCE

Product Engineer — Associate | Gowra Aerospace Technologies Pvt. Ltd. — India May 2022 – Jul 2022 (*Internship*)

- Supported aerospace manufacturing programs from raw material processing through final component delivery, ensuring compliance with aerospace quality standards while coordinating engineering documentation across production and quality teams.
- Performed dimensional inspection and tolerance analysis using Geometric Dimensioning and Tolerancing (GD&T); applied Failure Mode and Effects Analysis (FMEA) and Statistical Process Control (SPC) to identify manufacturing risks, improve process reliability, and validate structural component geometry.

Graduate Engineering Trainee — Brahmastra Aerospace Pvt. Ltd. — India Jul 2023 – Sep 2023 (*Internship*)

- Designed aerospace components using CATIA and SolidWorks (parametric and surface modeling); generated engineering drawings and technical documentation in compliance with GD&T standards.
- Performed structural finite element analysis (FEA) using ANSYS to evaluate stress distribution and structural response under operational loading, supporting design validation and engineering decision-making.

EDUCATION

Master of Science — Aerospace Engineering | Auburn University (R1 Research Institution) — Auburn, USA (Graduated May 2026)

Relevant Coursework: Fatigue & Damage Tolerance, Composite Materials, Experimental Mechanics, Viscoelasticity, Quantum Physics, Lightweight Space Structures, Deformation and Failure of Engineering Materials, Biomaterials, Phase Transformations in Materials, Fluid Dynamics

Bachelor of Technology — Aeronautical Engineering | Manipal Institute of Technology — India (Graduated 2023)

Relevant Coursework: Aircraft Design, Structural Mechanics, Finite Element Methods, Propulsion, Aerodynamics, Aerospace Materials

TECHNICAL PROJECTS

Composite Laminate Analysis Toolkit — Python | [GitHub](#)

- Implemented Classical Lamination Theory (Python/NumPy, Streamlit): per-ply transformed stiffness, A/B/D matrices, and ply stresses and strains in both laminate and material axes for user-defined stacking sequences, with layup visualization and one-click PDF report generation.
- Built a ply-discount progressive failure solver (Maximum Stress, Tsai-Hill, Tsai-Wu) that degrades matrix- and fibre-failed plies and re-solves the laminate to report first-ply and last-ply load factors (2.9× and 5.0× for a T300/5208 [0/45/-45/90]_s laminate at $N_x = 100$ N/mm); added a layup optimizer searching 256 symmetric [a/b/c/d]_s candidates.
- Validated against published textbook examples (Kaw): laminate stiffness within 0.5%, Tsai-Wu first-ply failure load within 0.05%, last-ply load exact; benchmarking exposed and corrected errors in failure-criteria stress axes and the Tsai-Hill cross term; 15 automated pytest tests.

Structural Integrity & Fatigue / Fracture Analysis Toolkit — Python | [GitHub](#)

- Developed a Python/Streamlit fatigue and fracture toolkit integrating Basquin S-N, Coffin-Manson strain-life, and Goodman/Gerber/Soderberg mean-stress diagrams (endurance strength from the Basquin line at 1e7 cycles) with a built-in database of 8 structural alloys.
- Implemented adaptive-step fatigue crack growth (Paris, Walker, Forman) with threshold non-propagation and a K_{max} -based fracture check, stress-intensity safety diagrams, rainflow counting with Miner's rule and Morrow/Smith-Watson-Topper mean-stress correction, and Monte Carlo reliability (up to 5,000 samples; median and 10th-percentile life).
- Validated against published examples: Paris-law life within 0.1% of a worked example (128,697 vs 128,792 cycles; SAE 1020 steel, 1 mm to 68 mm edge crack) and rainflow output identical to the ASTM E1049 example; benchmarking corrected Basquin amplitude/2N and mean-stress errors; 23 automated pytest tests.

Hypersonic Missile Flow Simulation & Gas Turbine Blade Modal Analysis — ANSYS Fluent

- Modeled external hypersonic flow over a missile at Mach 10, 90 km altitude, and 5° angle of attack in ANSYS Fluent using the k- ω turbulence model and pressure-far-field boundary conditions, converging lift and drag coefficients within 100 iterations.
- Generated pressure, velocity-Mach, and shock-cone contour visualizations to characterize aerodynamic loading, shock structure, and boundary-layer behavior across the missile surface and control fins.
- Performed modal analysis of a Titanium-alloy high-pressure turbine (HPT) blade (NACA 2412 profile, 250 mm blade length) under 700 N tangential load, 400 N axial load, and 4,252 RPM rotational velocity; extracted the first five natural vibration mode shapes to assess resonance risk.

Aging Effects in Magnetorheological Elastomers — MATLAB / ANSYS

- Fabricated silicone-based MR elastomer samples (with and without 27 vol.% carbonyl-iron particles) and designed a CATIA/AutoCAD test fixture, waterjet-CNC machined in-house, to mount specimens for free-vibration damping tests.
- Quantified humidity-driven aging effects via LabVIEW-based accelerometer data acquisition and logarithmic-decrement analysis, measuring a 15–21% increase in damping ratio and up to a 20% reduction in natural frequency after 2 weeks of environmental-chamber exposure.

CERTIFICATIONS

Google Project Management Professional Certificate — Google

Infor ERPLN — LN Engineering Module (Pursuing Certification), via coursework